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CFD SIMULATION OF UNSTEADY 2D HEAT DIFFUSION ON A SQUARE PLATE USING VARIOUS TIME STEPPING SCHEMES

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ABSTRACT
This article studies the time evolution of the 2D heat diffusion equation with a heat source perpendicular to the plane of the domain. This has many applications such as a LASER beam or an electron beam irradiating onto a thin metal plate for material processing. The 2D unsteady heat diffusion equation is solved numerically using the Finite Volume Method (FVM) in the C++ programming language.

NOMENCLATURE
T Temperature
ρ Density
C_p Specific Heat
k Thermal Conductivity
α Thermal Diffusivity
q Source Term
Δv Volume of an element
Δt Time Step
Fo Fourier Number

1 INTRODUCTION
Material transformation processes involving energetic beams such as LASER hardening, LASER beam welding, electron beam welding, LASER Selective Melting, etc. are a hot topic of research for the past two to three decades. There are many experiments done by changing the type of the material[1][2][3], type of the beam[4][5] or parameters of the beam[6][7] to understand the influence of such processes on the microstructure of the metal. Numerical studies of such processes also have been proven to be useful in understanding the physics of such problems.
A numerical model using FVM has been developed in C++ to study the heat diffusion in the square plate with a localized source term. Three different time stepping schemes have been used - the explicit, implicit and the Crank-Nicholson scheme to compare their performance against each other. The results have been validated with simulation results in COMSOL Multiphysics software.

2 NUMERICAL METHODS
2.1 Governing Equation
The electron beam or LASER beam can be considered as a localized heat source (either stationary or moving depending on the type of analysis) acting perpendicular to the plane of the square plate. The heat conduction in the third direction is negligible if the plate is very thin and in that case, it can be solved as a 2D problem. The governing equation is the 2D unsteady heat diffusion equation given in Equation (1).

$$\rho C_p \frac{\partial T}{\partial t} = \nabla.(k\nabla T) + q \tag{1}$$

To employ the FVM, we discretize the governing equation. We start by integrating Equation (1) over the control volume and for a time step dt such that the limits are from t to $t+dt$ (Equation

(2)).

$$\int_t^{t+dt}\int_{CV}\rho C_p\frac{\partial T}{\partial t}dvdt=\int_t^{t+dt}\int_{CV}(\nabla.(k\nabla T)+q)dvdt \quad (2)$$

By using Gauss Divergence Theorem for the diffusion term, we get Equation (3).

$$\int_t^{t+dt}\int_{CV}\rho C_p\frac{\partial T}{\partial t}dvdt=\int_t^{t+dt}\sum_f(k\nabla T)_f.\bar{A}_f+\int_t^{t+dt}qdvdt \quad (3)$$

We make a linear profile assumption for the diffusion term with time so that

$$\int_t^{t+dt}\sum_f(k\nabla T)_f\cdot\bar{A}_f=g\left(\sum_f(k\nabla T)_f\cdot\bar{A}_f\right)^0 + (1-g)\left(\sum_f(k\nabla T)_f\cdot\bar{A}_f\right)^1\Delta t \quad (4)$$

where the 0 superscript refers to the values in the previous time step and 1 represents the values in the current time step. Substituting the expression for the diffusion term from Equation (4) back in Equation (3) and simplifying it for a particular finite volume with respect to the cell centroid and neighbouring values, we get

$$a_pT_p^1=\sum_{nb}a_{nb}\left(gT_{nb}^1+(1-g)T_{nb}^0\right) + T_p^0\left(a_p^0+(1-g)\left(q_p^0\Delta v-\sum a_{nb}\right)\right)+b \quad (5)$$

where $a_p=a_p^0+g(\sum_{nb}a_{nb}-q_p\Delta v)$, $a_p^0=\frac{\rho C_p\Delta v}{\Delta t}$ and $b=gq_c^1\Delta v+(1-g)q_c^0\Delta v$. In Equation (5), the subscript p denotes the centroid values, nb denotes the neighbouring values (east, west, north and south) and g gives various time stepping schemes. Putting $g=0$ simplifies the Equation (5) to an explicit scheme as the current value of temperature depends only the previous time step values. Similarly, putting $g=1$ gives the implicit scheme and $g=\frac{1}{2}$ gives the Crank-Nicholson scheme. By putting these values in Equation (5), rearranging them and simplifying, we get Equations (6), (7) and (8) for the explicit, implicit and Crank-Nicholson schemes respectively. In our case, $q_c=q$ and $q_p=0$, these are the coefficients that arise when the source term is linearized.

$$T_p^1=Fo\sum_{nb}T_{nb}^0+(1-4Fo)T_p^0+\frac{q\Delta t}{\rho C_p} \quad (6)$$

$$(1+4Fo)T_p^1=Fo\sum_{nb}T_{nb}^1+T_p^0+\frac{q\Delta t}{\rho C_p} \quad (7)$$

$$(1+2Fo)T_p^1=\frac{Fo}{2}(\sum_{nb}T_{nb}^0+\sum_{nb}T_{nb}^1)+(1-2Fo)T_p^0+\frac{q\Delta t}{\rho C_p} \quad (8)$$

In Equations (6), (7) and (8), all the material properties are assumed to be homogeneous, isotropic and constant. So, the value of density, thermal conductivity and specific heat is taken to be 1 in their respective units.

2.2 Boundary Conditions

All the boundary conditions are taken to be of Dirichlet type in this problem. Since the domain is a square plate of size 1x1, the temperature at all four edges (top, bottom ,left and right) is equal to some ambient temperature $T_a=298K$. In mathematical form,

$$\begin{aligned}T(x=0,y,t)&=T_a\\T(x=L,y,t)&=T_a\\T(x,y=0,t)&=T_a\\T(x,y=L,t)&=T_a\end{aligned} \quad (9)$$

The initial temperature in the entire domain is also taken to be equal to Ta.

$$T(x,y,t=0)=Ta \quad (10)$$

A top-hat beam distribution is assumed for the localized heat source which means that the value of the heat source is a constant q_0 in the area over which the beam is focused and zero elsewhere.

$$\begin{aligned}q(x,y)&=q_0 \quad \text{if } x_1<x<x_2 \text{ and } y_1<y<y_2\\&=0 \text{ everywhere else}\end{aligned} \quad (11)$$

where $x_1=\frac{L}{2}-2.5\Delta x$, $x_2=\frac{L}{2}+2.5\Delta x$, $y_1=\frac{L}{2}-2.5\Delta y$ and $y_2=\frac{L}{2}+2.5\Delta y$. The boundary conditions and heat source details are better understood from Figure 1.

2.3 Mesh

A uniform orthogonal structured mesh with square elements is considered here. The Von Neumann stability criteria is used to determine the element size and time step size. Accordingly, the element size and time step are chosen: $dt=0.0001$ and $dx=0.05$ for $\alpha=1$. The 1x1 square domain is meshed into 20x20 elements as illustrated in Figure 2. Since a square meshed is used here, $dy=dx=0.05$ and $dv=dxdy=(dx)^2$.

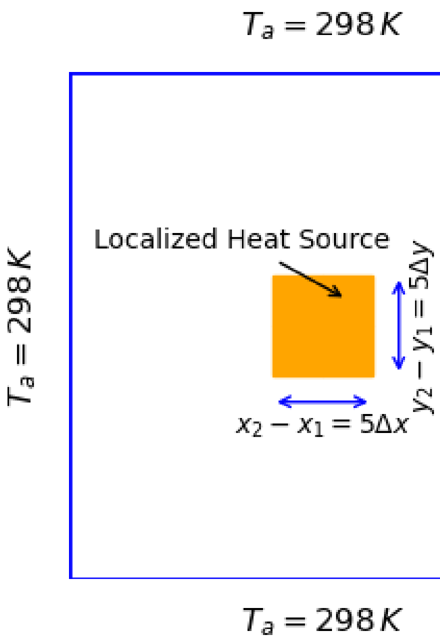

FIGURE 1: THE DOMAIN WITH BOUNDARY CONDITIONS AND LOCALIZED HEAT SOURCE.

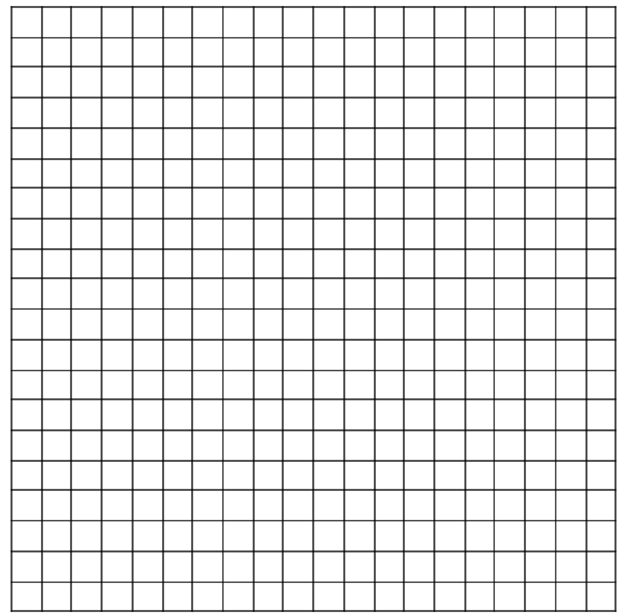

FIGURE 2: 20x20 UNIFORM ORTHOGONAL AND STRUCTURED MESH FOR THE SQUARE PLATE.

3 RESULTS AND DISCUSSION

Equations (6), (7) and (8) are solved using the FVM code prepared in C++. For the implicit and Crank Nicholson time stepping, the values of the temperature at the current time step are required at every time step. They are found out by using Line-By-Line Tri-Diagonal Matrix Algorithm (LBL-TDMA) at every time step. Because LBL-TDMA is an iterative method, the equations need to converge in space at every time step. The tolerance for convergence in space is $1e^{-6}$ and for convergence in time is $1e^{-4}$. All three simulations were run for a 2000 time steps which is a time of $2000\Delta t = 0.2s$ by which time the system reached steady state. The temperature distribution of $t = 0s$, $t = 0.05s$ and $t = 0.2s$(steady state) is presented in Figure 3.
We can see that at $t = 0$ s, the temperature is 298 K everywhere according to the initial condition. At $t = 0.05$ s, the heat due to the localized heat source has dissipated a little bit and the temperature at the center is around 300 K while at $t = 0.2$ s, the heat has dissipated completely and the maximum temperature is the temperature at the center which is around 301.5 K. The results shown in Figure 3 are for the explicit scheme. To compare the performance of the three schemes, the number of iterations required for convergence and the error of each scheme is summarized in Table 1 for $\Delta t = 0.0001s$.

Scheme	No. of iterations	%Error
Explicit	1949	0.04198
Implicit	1993	0.04331
CN	1938	0.04132

TABLE 1: NUMBER OF ITERATIONS REQUIRED IN EACH SCHEME FOR CONVERGENCE FOR $\Delta t = 0.0001s$

Scheme	No. of iterations	%Error
Explicit	1152	0.027526
Implicit	1185	0.026531
CN	1106	0.029184

TABLE 2: NUMBER OF ITERATIONS REQUIRED IN EACH SCHEME FOR CONVERGENCE FOR $\Delta t = 0.0002s$

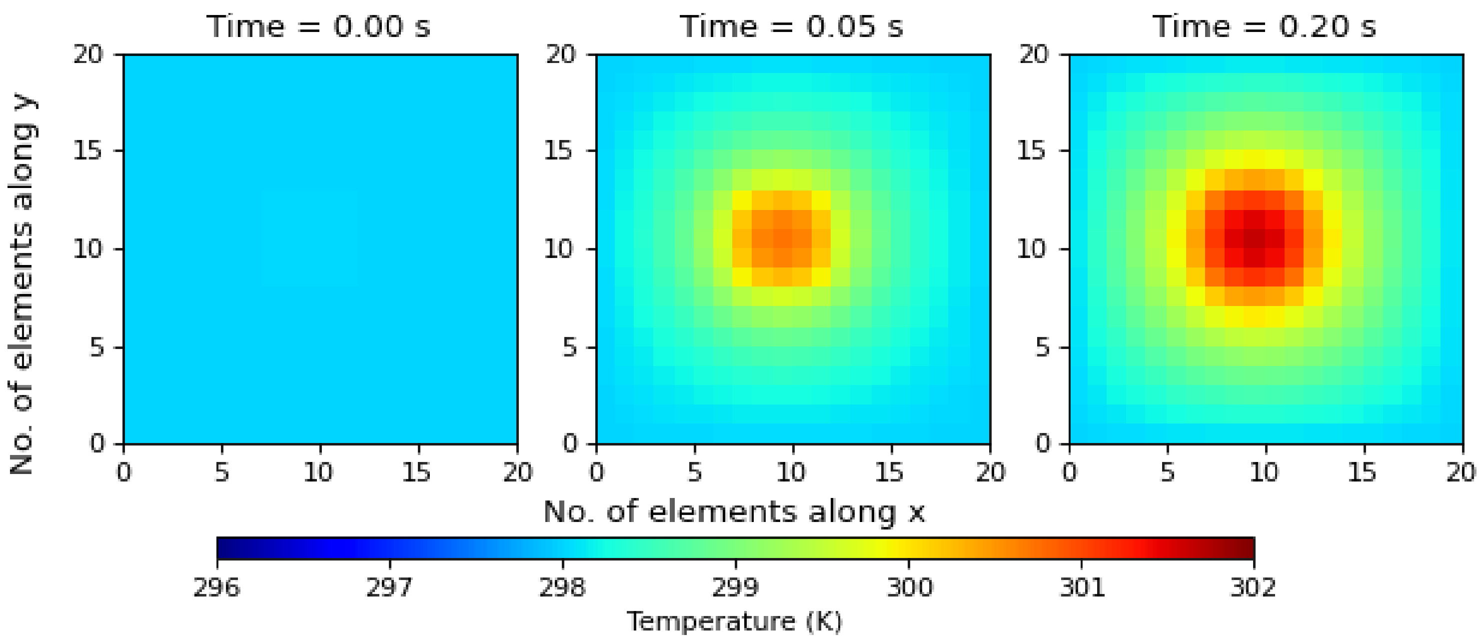

FIGURE 3: TEMPERATURE DISTRIBUTION FOR VARIOUS TIMES.

Scheme	No. of iterations	%Error
Explicit	718	0.026862
Implicit	721	0.027526
CN	647	0.026531

TABLE 3: NUMBER OF ITERATIONS REQUIRED IN EACH SCHEME FOR CONVERGENCE FOR $\Delta t = 0.000375s$

The same exercise is done for $\Delta t = 0.0002s$ and $\Delta t = 0.000375s$ and the results are summarized in Tables 2 and 3 respectively. We observe from Tables 1, 2 and 3 that as the time step size increases or the mesh in time becomes coarser, the number of iterations in time required for convergence to the steady state reduces. This makes sense because larger step size means faster convergence but at a loss of accuracy. Step size of $\Delta t = 0.000375s$ is the maximum possible step size that satisfies the stability analysis. The percentage error of each method is calculated after 2000 iterations (i.e. when the system has reached steady state in all cases) with reference to the values obtained from the COMSOL Multiphysics commercial software which employs the Finite Element Method (FEM) to discretize the gov

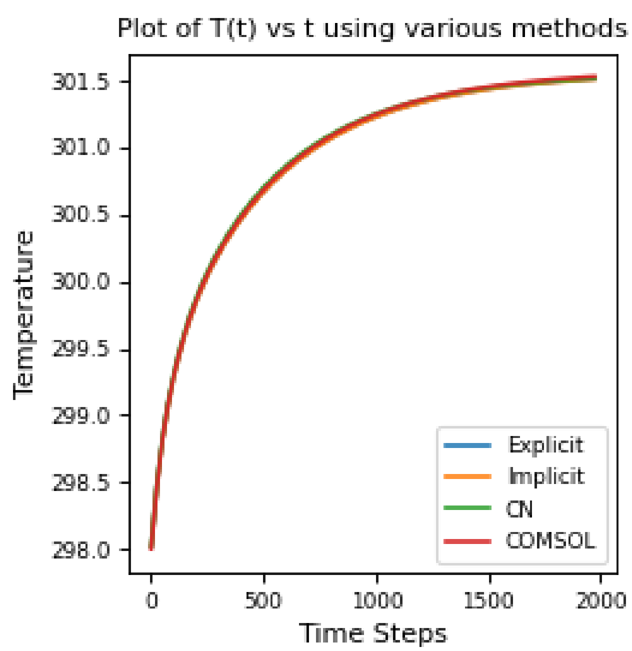

FIGURE 4: $T(t)$ for $\Delta t = 0.0001$.

-erning equation. Same mesh parameters were used when simulating in COMSOL also. It is also observed that the Crank Nicholson scheme experiences the fastest convergence followed by the explicit and implicit schemes respectively for all three time steps considered. It is difficult to comment on the accuracy of the solution because the validation is done by taking the same mesh.

To further understand the relative performance of the methods, the variation of temperature at the center of the plate as a function of time for all three schemes along with COMSOL results is presented in Figure 4. It shows that there is excellent agreement between the three schemes and the COMSOL result. Also, the results obtained for the three schemes are very close amongst each other. This is clear from the Tables 1, 2 and 3 also.

The C++ code and the validation input file of COMSOL along with all the plots have been provided as supplementary data along with this article.

4 CONCLUSIONS

A numerical solver for 2D unsteady heat diffusion is developed with FVM discretization using the explicit, implicit and Crank-Nicholson time stepping schemes. The Crank-Nicolson method is found to converge the fastest followed by the explicit and implicit schemes respectively. Validation is done with results from COMSOL simulation with same mesh parameters and the results are in excellent agreement with the COMSOL results. This solver can now be used for understanding beam-material interactions, even if the beam is moving. The code already contains the velocity initialization. For the purpose of validation, the velocity was taken to be zero in this case. With slight modification to the code, a finite value of velocity can be given to understand the temperature distribution due to a moving heat source which is more useful practically as the beam usually traverses in a particular direction on the material surface.

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REFERENCES

[1] Salleh MN, Ishak M, Aiman MH, Zaifuddin Q, Quazi MM. The effect of laser surface hardening on the surface hardness of mild steel. *IOP Conference Series: Materials Science and Engineering*. 2020;788(1):012014.

[2] Lu Y, Ehle LC, Richter S, Radel T. Influence of multi-pass laser hardening of normalized AISI 4140 on the grain size. *Surface and Coatings Technology*. 2021;421:127434.

[3] Muthukumaran G, Dinesh Babu P. Laser transformation hardening of various steel grades using different laser types. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*. 2021;43:1–29.

[4] Pashby IR, Barnes S, Bryden BG. Surface hardening of steel using a high power diode laser. *Journal of Materials Processing Technology*. 2003;139(1–3):585–588.

[5] Ganeev RA. Low-power laser hardening of steels. *Journal of Materials Processing Technology*. 2002;121(2–3):414–419.

[6] Nath AK, Gupta A, Benny F. Theoretical and experimental study on laser surface hardening by repetitive laser pulses. *Surface and Coatings Technology*. 2012;206(8–9):2602–2615.

[7] Maharjan N, Zhou W, Zhou Y, Guan Y, Wu N. Comparative study of laser surface hardening of 50CrMo4 steel using continuous-wave laser and pulsed lasers with ms, ns, ps and fs pulse duration. *Surface and Coatings Technology*. 2019;366:311–320.